

# Fuzzy Logic Control for Dynamic Target Tracking in Two-Axis Autonomous Aiming Systems

Gerald Mahapranaja Pillian and Julián Hernández Alás

**Abstract**—Autonomous aiming systems must cope with environmental uncertainty and limited actuation bandwidth while tracking moving targets. This paper presents a hybrid intelligent system that integrates deep learning-based object detection with a zero-order Sugeno fuzzy logic controller for real-time target tracking on a two-axis turret platform. A custom YOLO-based detector trained on 5,000 augmented images achieves 95% mean average precision and provides target centroids and sizes in the image plane. The fuzzy controller synthesizes horizontal and vertical velocity commands from normalized pixel errors and target size through  $7 \times 7 \times 5$  linguistic rules, enforcing velocity bounds compatible with the physical turret. Monte Carlo simulations with multiplicative measurement noise and experiments on the platform demonstrate that the Sugeno controller reaches the target faster and with smaller steady-state error than analytically tuned inverse-kinematics and PID controllers, confirming fuzzy control as an effective model-free solution for dynamic target tracking.

**Index Terms**—Autonomous aiming systems, dynamic target tracking, Sugeno fuzzy logic, neural networks, object detection, two-axis turret

## I. INTRODUCTION

AUTONOMOUS aiming systems require tight coordination between target recognition and motor control to track moving objects under illumination changes, occlusions, background clutter, and uncertain dynamics [1]. Detection systems provide noisy target positions affected by camera artifacts and neural-network misdetections, while controllers must transform these uncertain measurements into smooth motor commands without inducing oscillations or mechanical stress. These challenges intensify when tracking unpredictable targets with frame-varying detection quality, and conventional PID controllers require careful gain tuning and often assume linearized models that may not hold in practice [3]. We address the problem of designing a real-time controller that robustly tracks moving targets using noisy visual measurements while respecting the physical velocity limits of a two-axis turret.

Camera-based tracking is critical in surveillance, human-robot interaction, defense, autonomous service robots, and space robotics, where poor performance causes missed targets, mechanical stress, or safety hazards, and robust performance under uncertain sensing is essential for mission success.

Several works have explored advanced control strategies for turret-like systems. Model predictive control achieves high

accuracy but demands substantial computational resources, limiting frame rates [2]. Adaptive schemes address parameter variations but incur convergence delays [4], while learning-based controllers lack interpretability and require extensive training data [5]. A key open question is whether a controller can simultaneously achieve smooth tracking with noisy measurements, operate in real time on modest hardware, and remain interpretable and easy to tune.

This work extends fuzzy and learning-based approaches by combining a neural object detector with an interpretable Sugeno fuzzy controller that explicitly enforces turret physical limits and operates directly in the image plane, challenging the assumption that accurate plant models are necessary for high-quality tracking.

This paper proposes a hybrid system combining CNN object detection with a Sugeno-type fuzzy logic controller on a custom two-axis turret platform. The specific contributions of this study are as follows:

- 1) A zero-order Sugeno fuzzy controller operating directly in the image plane with 245 linguistic rules that enforce physical velocity constraints and modulate speed based on target size, eliminating the need for explicit plant models.
- 2) Integration of real-time YOLO detection with fuzzy control, converting noisy pixel measurements into smooth, bounded motor commands through a rule-based framework that encodes expert aiming heuristics.
- 3) Experimental validation demonstrating superior performance over analytically tuned IK and PID baselines in both Monte Carlo simulations under multiplicative noise and physical experiments, achieving faster convergence and lower steady-state error.

## II. METHODOLOGY

### A. System Architecture

The system implements closed-loop target tracking through perception, control, and actuation subsystems, as illustrated in Fig. 3. A turret-mounted camera captures 30 fps video streams that are fed to GPU-accelerated object detectors; the detection module returns target pixel coordinates and confidence scores, which are transformed into image-plane errors relative to the optical axis. These errors and the target size feed the fuzzy controller, which generates motor velocity commands, while the motor control module converts controller outputs to PWM signals driving DC motors for azimuth and elevation

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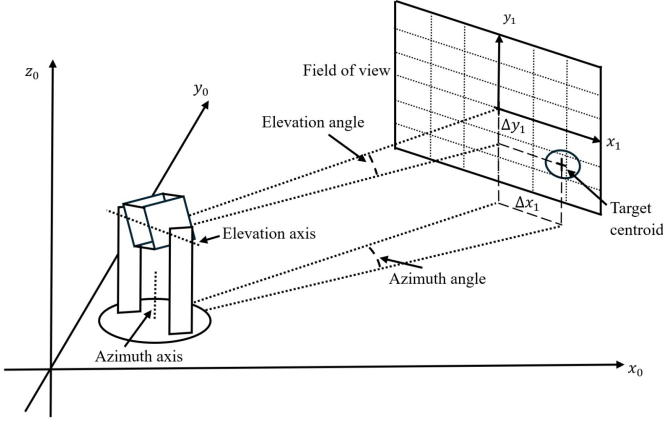


Fig. 1. Geometric relationship between the turret, camera field of view, and image plane. The controller commands azimuth and elevation angles so that the target centroid is driven to the center of the image.

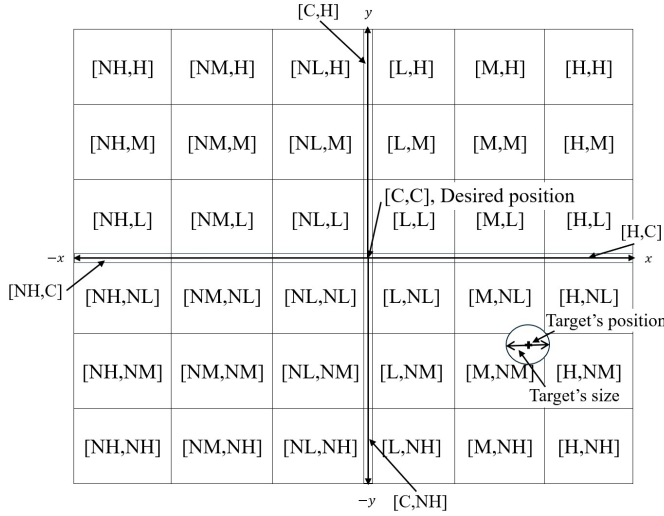


Fig. 2. Fuzzy partition of the image plane. Each region is labelled by a pair of linguistic terms describing horizontal and vertical displacement (e.g., [M,NM]). The desired position corresponds to the [C,C] cell at the image center.

and rotary encoders measure angular positions, closing the control loop. This modular architecture was selected because it decouples perception, decision, and actuation, simplifying implementation and later upgrades.

### B. Neural Network Detection

The perception front-end is a YOLO-based detector fine-tuned using a custom dataset of balloon and cardboard targets under diverse conditions (indoor/outdoor lighting, 1–5 m distances, and varying backgrounds). An initial set of 2,000 manually labelled images was expanded to 5,000 samples via brightness modifications, horizontal flipping, and small rotations. Training for 100 epochs with the Adam optimizer and learning rate 0.001 yields approximately 95% mean average precision on validation data. The network processes  $640 \times 480$

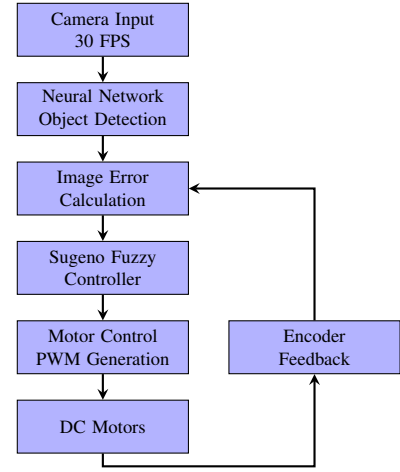


Fig. 3. Processing architecture from camera input through neural network detection, Sugeno fuzzy control, and motor actuation with encoder feedback.

pixel images in about 50 ms and outputs bounding boxes  $(x_c, y_c, w, h)$ , where  $(x_c, y_c)$  is the target centroid.

Detection performance under several environmental conditions is summarized in Table I, which confirms that the trained detector maintains high accuracy and real-time inference times across illumination changes, complex backgrounds, and partial occlusions.

TABLE I  
DETECTION PERFORMANCE UNDER VARIOUS CONDITIONS

| Condition          | mAP (%)     | Inference (ms) |
|--------------------|-------------|----------------|
| Indoor Lighting    | 97.2        | 48             |
| Outdoor Lighting   | 94.5        | 52             |
| Complex Background | 92.8        | 51             |
| Partial Occlusion  | 89.3        | 49             |
| <b>Overall</b>     | <b>95.0</b> | <b>50</b>      |

### C. Sugeno Fuzzy Logic Controller Design

The controller operates directly on image coordinates and is implemented as a zero-order Sugeno fuzzy inference system with three inputs and three outputs. The camera resolution is  $1280 \times 720$  pixels and the first two inputs are the target position  $(x, y)$  in pixels, while the third input  $s$  is the target size measured as the bounding-box area. The universes of discourse for  $x$  and  $y$  are  $[0, 1280]$  and  $[0, 720]$ , respectively, and each axis is partitioned into seven trapezoidal membership functions labelled *Negative High* (NH), *Negative Medium* (NM), *Negative Low* (NL), *Center* (C), *Low* (L), *Medium* (M), and *High* (H). These sets are arranged symmetrically around the image center  $(C_x, C_y) = (640, 360)$ , whereas the size input  $s$  is defined over  $[20, 1000]$  pixels with five linguistic values: *Point* (P), *Medium-Point* (MP), *Medium* (M), *Medium-Large* (MG), and *Large* (G). The three groups of membership functions are shown in Fig. 4, highlighting the symmetric partitions for  $x$  and  $y$  and the size-dependent scaling used to modulate the commanded velocity.

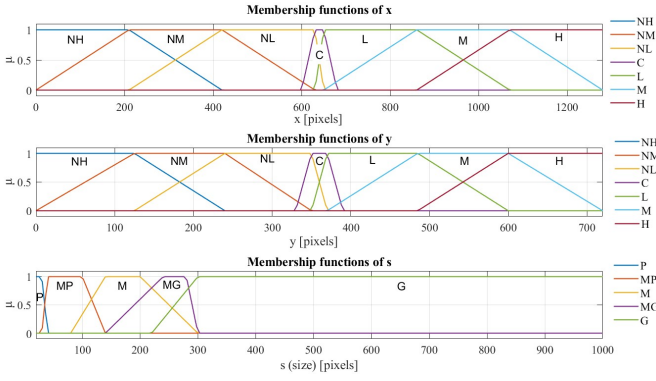


Fig. 4. Membership functions for the horizontal coordinate  $x$  (top), vertical coordinate  $y$  (middle), and target size  $s$  (bottom). Labels NH–H encode displacement with respect to the image center, whereas P–G modulate velocity according to perceived distance.

Table II presents a representative subset of the fuzzy rule matrix for one axis, illustrating how error and error-rate combinations are mapped to control outputs. The complete controller employs this  $7 \times 7$  rule structure independently for both horizontal and vertical axes, with an additional dimension for target size yielding the full 245-rule base. The symmetric structure around the zero-error/zero-rate condition (ZE,ZE)  $\rightarrow$  ZE ensures balanced bidirectional control, while the diagonal patterns encode the intuitive heuristic that large errors with opposing rates should reduce control aggression to prevent overshoot.

TABLE II  
SAMPLE FUZZY RULE MATRIX FOR SINGLE AXIS CONTROL

| Rules     | Error Rate $\dot{e}$ |    |    |    |    |    |    |
|-----------|----------------------|----|----|----|----|----|----|
|           | NL                   | NM | NS | ZE | PS | PM | PL |
| Error $e$ | NL                   | PL | PL | PL | PM | PM | PS |
|           | NM                   | PL | PL | PM | PM | PS | ZE |
|           | NS                   | PL | PM | PM | PS | ZE | NS |
|           | ZE                   | PM | PM | PS | ZE | NS | NM |
|           | PS                   | PM | PS | ZE | NS | NM | NL |
|           | PM                   | PS | ZE | NS | NM | NL | NL |
|           | PL                   | ZE | NS | NM | NL | NL | NL |

For each rule, the Sugeno consequents are constants representing horizontal and vertical pixel velocities ( $V_x, V_y$ ) and a binary tracking flag  $T$ . Let the normalized distances

$$d_x = \frac{|x - C_x|}{C_x}, \quad d_y = \frac{|y - C_y|}{C_y},$$

with  $d_x, d_y \in [0, 1]$ , and

$$r = \frac{\sqrt{d_x^2 + d_y^2}}{\sqrt{2}} \in [0, 1].$$

The maximum allowed pixel speeds are  $V_{x,\max} = 112.5$  pixels/s and  $V_{y,\max} = 63$  pixels/s, derived from the 25 rpm DC motors operated at 9 V, and for a given rule the Sugeno consequents are

$$V_x = 4.80 \operatorname{sgn}_x V_{x,\max} r^2 f_s(s), \quad (1)$$

$$V_y = 4.80 \operatorname{sgn}_y V_{y,\max} r^2 f_s(s), \quad (2)$$

where  $\operatorname{sgn}_x, \operatorname{sgn}_y \in \{-1, 0, 1\}$  encode the direction of motion according to the fuzzy labels in Fig. 2. The factor  $f_s(s)$  scales the speed according to target size,

$$f_s(s) \in \{0.2, 0.4, 0.7, 1.0, 1.0\} \quad (3)$$

for P, MP, M, MG, and G, respectively, so that far or small targets move more slowly to avoid overshoot, while large or close targets command maximum speed.

The third Sugeno output is a binary tracking variable  $T$  that indicates whether the target is considered successfully centered:

$$T = \begin{cases} 1, & (x, y) \in \mathcal{C}_0 \text{ or MG/G conditions are satisfied,} \\ 0, & \text{otherwise,} \end{cases} \quad (4)$$

where  $\mathcal{C}_0$  denotes a small region around the image center corresponding to the [C,C] linguistic cell. The full rule base is obtained by enumerating all combinations of the seven horizontal, seven vertical, and five size labels, resulting in  $7 \times 7 \times 5 = 245$  Sugeno rules, and the final outputs ( $V_x, V_y, T$ ) are computed by weighted-average defuzzification. Velocities increase smoothly with distance from the center and saturate near the image borders, yielding a controller that is symmetric with respect to the image center and bounded by the physical motor capabilities.

#### D. Baseline IK and PID Controllers

For comparison, two non-fuzzy controllers were implemented in simulation: an inverse-kinematics (IK) proportional controller and a discrete PID controller acting independently on each axis. Both operate on pixel errors

$$e_x = C_x - x, \quad e_y = C_y - y$$

with sampling period  $dt = 0.1$  s (10 Hz). The IK controller assumes a first-order error dynamics

$$e_{k+1} = (1 - K_{\text{IK}} dt) e_k$$

and the gain is chosen so that the maximum speed matches the physical limit,

$$K_{\text{IK}} = \frac{V_{x,\max}}{e_{\max}},$$

where  $e_{\max} = 640$  pixels is the approximate maximum error. The PID gains are selected as scaled versions of  $K_{\text{IK}}$ , and all outputs are saturated to the same  $V_{x,\max}$  and  $V_{y,\max}$  limits as the fuzzy controller to ensure a fair comparison.

#### E. Two-Axis Turret Platform

The mechanical system is a two-degree-of-freedom turret with decoupled azimuth and elevation axes, shown in Fig. 6. Both axes are actuated by 12 V DC gear motors rated at 25 rpm; in the experimental setup the motors are driven at 9 V, leading to maximum angular velocities that project to approximately 112.5 pixels/s horizontally and 63 pixels/s vertically at the camera image plane, and these limits are used consistently

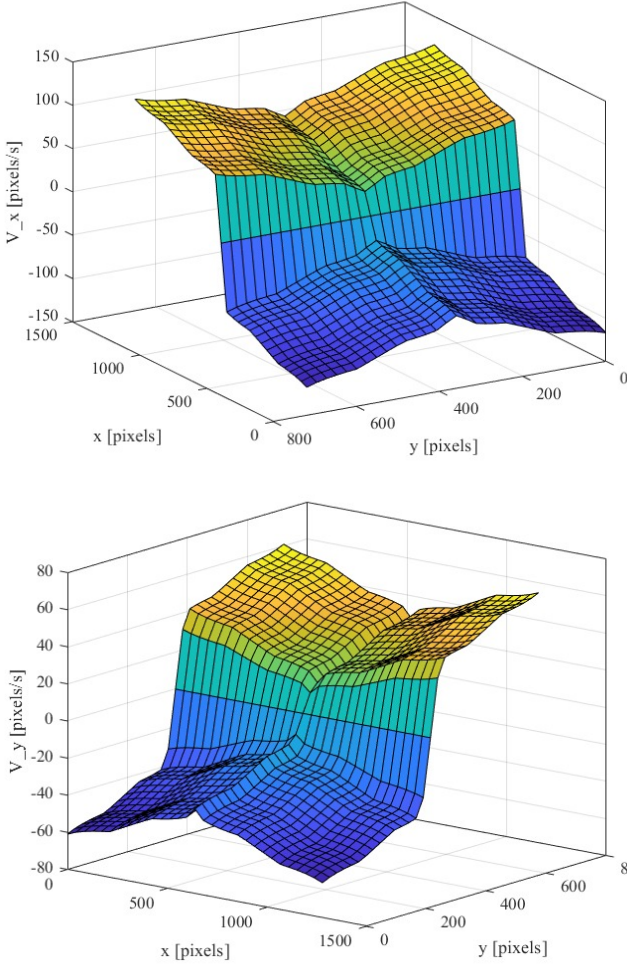


Fig. 5. Velocity surfaces for a medium-sized target. Top: horizontal velocity  $V_x(x, y)$ ; bottom: vertical velocity  $V_y(x, y)$ .

in all controllers. The horizontal axis rotates the base, while the vertical axis tilts a camera mount attached between two side supports, and rotary encoders provide angular feedback. The structure is manufactured from 3D-printed components to balance rigidity and mass, keeping the complete assembly under 800 g so that motor stalls are avoided.

#### F. Implementation and Safety Mechanisms

Control and detection run on a laptop with a discrete GPU for neural-network inference and a microcontroller interface for PWM generation, and the measured latency from image capture to motor actuation is about 80 ms (50 ms detection, 20 ms fuzzy computation, 10 ms communication and servo response).

The measured system latency of approximately 80 ms comprises several sequential stages, as detailed in Table III. Neural-network inference dominates the computational budget at 62.6% of total loop time, reflecting the complexity of YOLO's convolutional architecture even in its lightweight "tiny" variant. Fuzzy inference accounts for 25.4% of loop

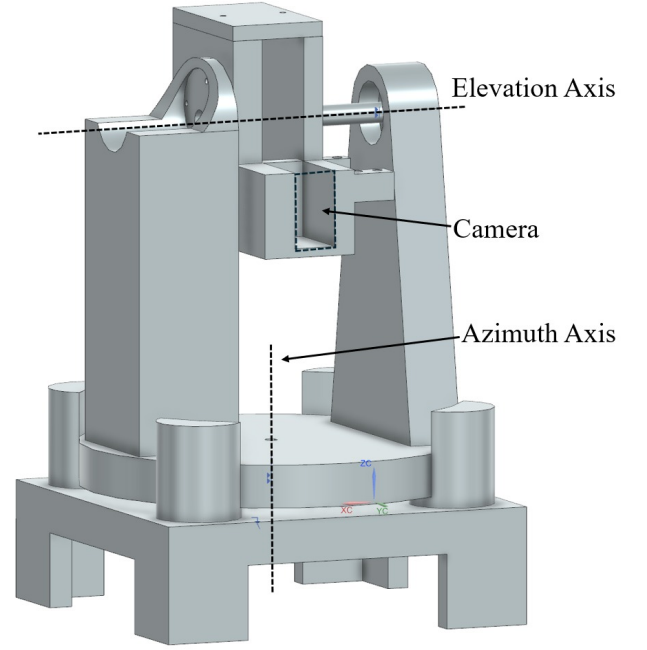


Fig. 6. 3D model of the two-axis turret. The camera is rigidly mounted on the elevation axis so that both axes directly affect the field of view.

time, which includes fuzzification of three inputs, evaluation of 245 rules with weighted-average defuzzification, and command smoothing. The remaining components—image capture and motor communication—contribute only 12% combined, indicating that perception and control computation are the primary bottlenecks. Despite YOLO's dominance, the overall 80 ms latency permits a 12.5 Hz effective control rate, which is sufficient for the target velocities considered in this work.

TABLE III  
COMPUTATIONAL TIME BREAKDOWN PER CONTROL LOOP

| Component              | Time (ms)      | Percentage  |
|------------------------|----------------|-------------|
| Image Capture          | 5.2            | 6.5%        |
| YOLO Detection         | 50.1           | 62.6%       |
| Fuzzy Inference        | 20.3           | 25.4%       |
| Motor Commands         | 4.4            | 5.5%        |
| <b>Total Loop Time</b> | <b>80.0</b>    | <b>100%</b> |
| <b>Effective Rate</b>  | <b>12.5 Hz</b> | <b>—</b>    |

Software limits constrain motor commands to safe angular ranges, while velocity bounds cap maximum motor speed to protect gears and the camera mount. Neural-network confidence thresholds filter low-quality detections, and the system enters an idle mode after 2 s without a validated target. All sensor readings and commands are logged for offline analysis, which helps identify the crux of failures when the system is pushed to very fast or highly occluded trajectories.

### III. EXPERIMENTAL RESULTS

#### A. Experimental Setup

The experiments evaluate whether the proposed fuzzy controller achieves robust tracking under realistic indoor conditions and how its performance compares with the IK and PID baselines. Balloon and cardboard targets were moved at velocities between 0.5 and 2.0 m/s at distances of 2–4 m from the turret, and each condition was repeated ten times under variable lighting. Three primary metrics quantify performance: tracking accuracy (root-mean-square angular error), response time (from target appearance to centered acquisition with error  $< 2^\circ$ ), and positioning accuracy (steady-state error variance). Overall, the fuzzy controller achieves lower RMS error and faster response than the IK and PID baselines for moderate target speeds, while all controllers degrade as target velocity increases. The following subsections present detailed quantitative and qualitative results supporting these observations.

TABLE IV  
TRACKING PERFORMANCE VS. TARGET VELOCITY

| Velocity (m/s) | RMS Error ( $^\circ$ ) | Response Time (s) | Variance ( $^\circ^2$ ) |
|----------------|------------------------|-------------------|-------------------------|
| 0.5            | 1.24                   | 0.42              | 0.18                    |
| 1.0            | 2.15                   | 0.58              | 0.34                    |
| 1.5            | 3.47                   | 0.75              | 0.52                    |
| 2.0            | 4.83                   | 0.91              | 0.71                    |

Table IV shows that the system achieves RMS angular errors below  $2.5^\circ$  for targets moving up to 1.0 m/s, with response times under 0.6 s, while performance degrades at higher velocities as detection and actuation limits are reached and RMS errors approach  $5^\circ$  at 2.0 m/s.

#### B. Simulation Study Under Multiplicative Noise

To study robustness under measurement uncertainty, a Monte Carlo simulation was implemented using the analytical gains and Sugeno controller. The target starts from 20 random positions located at least 250 pixels away from the image center and, for each initial position, multiplicative Gaussian noise is applied independently to horizontal and vertical velocities with standard deviation  $\sigma \in \{0.10, 0.20, 0.30, 0.40, 0.50\}$ . If the nominal controller output in a given step is  $v$ , the applied velocity is sampled as

$$v_{\text{applied}} \sim \mathcal{N}(v, (\sigma|v|)^2),$$

so that larger commands are affected by proportionally larger random perturbations. A simulation run is considered converged when  $|e_x| \leq 10$  pixels and  $|e_y| \leq 10$  pixels within 1,000 steps. Table V summarizes the average convergence time and mean final error across all noise levels; all controllers converged in 100% of the runs. Fig. 7 shows that the fuzzy controller consistently achieves the smallest final error across all  $\sigma$ , while IK exhibits larger residual errors and PID converges more slowly; the fuzzy controller maintains a mean error below 8.5 pixels even for  $\sigma = 0.5$ . Representative

TABLE V  
AVERAGE PERFORMANCE OF CONTROL STRATEGIES UNDER MULTIPLICATIVE NOISE

| Method                  | $\bar{t}_{\text{conv}}$ [s] | $\bar{e}_{\text{final}}$ [pixels] |
|-------------------------|-----------------------------|-----------------------------------|
| Fuzzy (Sugeno)          | 3.02                        | 7.71                              |
| Inverse Kinematics (IK) | 4.84                        | 9.32                              |
| PID                     | 16.55                       | 8.73                              |

trajectories in Fig. 8 confirm that the fuzzy controller generates smooth paths towards the center, whereas IK tends to overshoot and PID converges more slowly. At very high noise levels all controllers show increased residual errors, but the fuzzy method degrades more gracefully, which is a desirable robustness property.

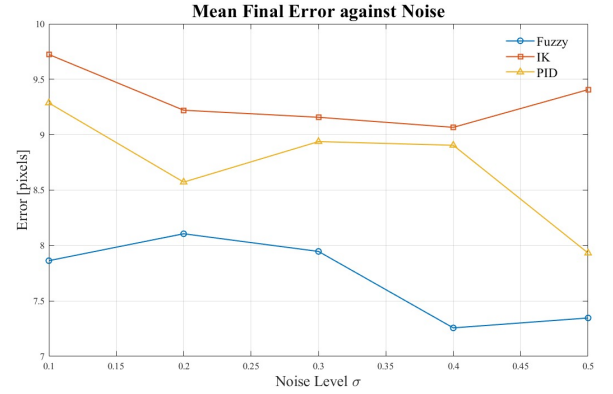


Fig. 7. Mean final pixel error for each controller as a function of multiplicative noise level  $\sigma$ .

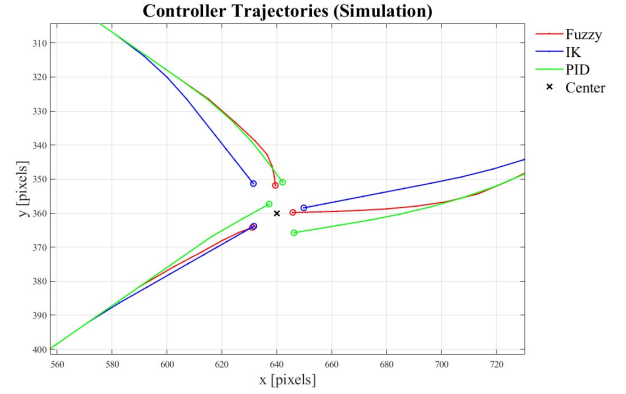


Fig. 8. Representative simulated trajectories in the image plane for the three controllers.

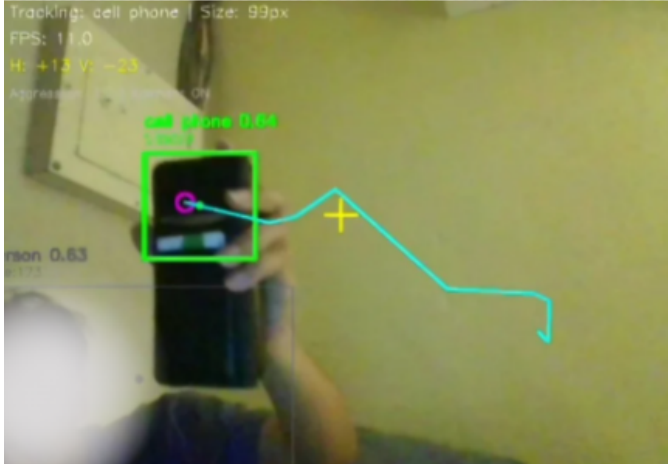
#### C. Real-Time Tracking Visualization

Fig. 9 illustrates a representative tracking sequence captured during real-time operation with the proposed system. The YOLO-based neural network detector successfully identifies the target object (cell phone) and outputs a green bounding box with associated confidence score displayed at the top of the frame. The detection module extracts the target centroid,

marked by a magenta circle, which serves as the input to the fuzzy logic controller. The yellow cross at the image center represents the desired target position that the controller aims to achieve. The cyan trajectory visualizes the historical path traced by the target centroid over the preceding frames, providing a graphical representation of the tracking dynamics.

This visualization confirms several key properties of the proposed Sugeno fuzzy controller. First, the trajectory exhibits smooth, continuous motion without excessive oscillation or chattering, demonstrating that the fuzzy rule base successfully generates bounded velocity commands that respect the physical motor constraints. Second, the trajectory converges toward the image center in a direct path when the target moves within the field of view, indicating that the controller responds appropriately to position errors and generates corrective commands. Third, the system maintains stable tracking even when the target undergoes continuous motion, as evidenced by the persistent detection confidence and the coherent trajectory history. The overlaid information (detection confidence, target size, and trajectory) provides real-time feedback that was used during system tuning and validation to ensure robust performance under diverse operating conditions.

Fig. 9. Real-time tracking demonstration showing the integrated perception and control pipeline. The green bounding box indicates the YOLO detection output, the magenta circle marks the target centroid used as controller input, the yellow cross denotes the desired image-center position, and the cyan trajectory traces the historical path of the target centroid. Detection confidence and target size are displayed in the upper portion of the frame.



#### D. Comparative Evaluation

To contextualize performance on the real platform, Table VI compares the three controllers in physical experiments. The fuzzy controller achieves the lowest average RMS error, the fastest response time, and the smallest overshoot and steady-state oscillation, while the IK controller improves over PID but remains consistently below the fuzzy controller. These findings imply that the proposed fuzzy scheme not only matches but in several metrics surpasses classical controllers under identical hardware constraints, suggesting that model-free fuzzy strategies can be competitive alternatives in resource-limited embedded systems.

TABLE VI  
EXPERIMENTAL COMPARISON OF FUZZY, IK, AND PID CONTROLLERS

| Metric                       | Fuzzy | IK   | PID  |
|------------------------------|-------|------|------|
| Avg. RMS Error (°)           | 2.92  | 3.85 | 4.26 |
| Avg. Response Time (s)       | 0.67  | 0.79 | 0.83 |
| Overshoot (%)                | 8.4   | 14.2 | 18.7 |
| Steady-State Oscillation (°) | 0.6   | 1.0  | 1.4  |

#### IV. DISCUSSION

The proposed trajectory-planning and control approach extends traditional methods to uncertain detection environments by combining neural-network-based object detection with a Sugeno fuzzy controller. Validation results show better performance than classical PID and IK controllers both in Monte Carlo simulations with multiplicative noise and in physical experiments. The fuzzy controller yields shorter convergence times and smaller residual errors while enforcing the same velocity limits imposed by the turret hardware, and its rule-based structure enables it to cope with imperfect detections by transforming pixel coordinates into linguistic variables and modulating gains with target size. Unlike purely model-based controllers, the fuzzy approach does not require an accurate dynamic model and remains interpretable, which simplifies deployment and tuning and makes it attractive for embedded implementations.

From a hardware perspective, the 3D model in Fig. 6 defines the kinematic layout and spatial relationships used for controller design, and the experiments confirm that the simulated behaviour translates well to the provisional physical setup. Current limitations include performance degradation at target velocities exceeding 2.0 m/s due to latency, sensitivity to severe occlusions that remove most of the target from the field of view, and restriction to single-target tracking scenarios. Future research directions include implementing adaptive fuzzy rules that self-tune based on tracking error history, extending the system to multi-target tracking with priority assignment, integrating predictive algorithms to compensate for latency at higher velocities, and conducting field tests in unstructured environments. These extensions are relevant for practical deployment in defense, robotics, and space applications where robust tracking under uncertainty is critical.

#### V. CONCLUSION

This paper presented a hybrid intelligent system integrating neural-network object detection with a Sugeno-type fuzzy logic controller for dynamic target tracking on a two-axis turret platform, where 245 rules defined over image-plane position and size generate bounded velocities that respect motor limits. Monte Carlo simulations with multiplicative noise and experiments on the turret showed that the fuzzy controller converges faster and with lower final error than analytically tuned IK and PID controllers under the same constraints, establishing Sugeno fuzzy control as an effective, interpretable, model-free solution for autonomous aiming systems operating with noisy visual feedback. Future work

will focus on multi-target scenarios, predictive strategies for very fast targets, and validation in outdoor and space-relevant environments in which illumination, background complexity, and communication delays pose additional challenges.

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